

ORIGINAL ARTICLE

Comparing maternal factors associated with postpartum depression between primiparous adolescents and adults: A large retrospective cohort study

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Abstract

Objectives: This study aimed to estimate the prevalence of diagnosed postpartum depression (PPD) and the likelihood of PPD among primiparous women. We also evaluated differences in the influence of various maternal factors associated with PPD in adolescent versus adult mothers.

Methods: We conducted a retrospective cohort study using electronic health records linked to birth certificates to evaluate the associations between maternal factors and PPD diagnosis. The study population was stratified into adults and adolescents based on age at delivery. We evaluated socioeconomic, demographic, psychological, and clinical factors associated with PPD in each of the age-defined maternal cohorts using multivariable logistic regression analyses.

Results: A total of 61,226 primiparous women, including 6435 (11%) mothers younger than 20 years old, were included in the study. The overall PPD rate was 4.0%, with the age-specific PPD rate measuring 1.6 times higher in adolescents than in adult women (6.1% vs. 3.8%). Compared with adults, adolescents were less likely to obtain first-trimester prenatal care (33% vs. 16%), more likely to have recent tobacco use (11% vs. 6%), and more likely to have had an infection during pregnancy (5% vs. 1%). In adjusted models, significant factors for PPD in both groups included a history of depression or anxiety, tobacco use, and long-acting reversible contraception use.

Conclusions: In this cohort of first-time mothers, adolescents had higher rates of PPD diagnosis as well as PPD-associated maternal factors than adults. Increased awareness of PPD risk in adolescents and early intervention, including integrating mental healthcare into prenatal care, may help benefit adolescents and reduce the risk and severity of PPD.

KEYWORDS

epidemiology, postpartum depression, risk factors, teenage pregnancy

1 | BACKGROUND

Postpartum depression (PPD) is defined as an episode of major depression associated with childbirth and parenting an infant.¹ According to a recent evidence review, the point prevalence of depression ranges from 4% to 34% during the first year postpartum in the US.² PPD is known to be associated with other adverse mental health outcomes for both mother and infant, including an increased suicide risk, poor mother-infant interactions, and an increased risk of child neglect, abuse, and homicide.^{3–5}

Postpartum depression is likely to result from the interplay of individual and societal influences. Prior research has identified a wide range of factors associated with the development of PPD, which can be broadly classified into demographic, socioeconomic, clinical, and psychological factors.^{6–8} Relevant studies, mostly carried out on adult women, suggest that women with lower socioeconomic status, single marital status, and those who are of Black, Hispanic, and Asian or Pacific Islander descent are at increased risk for PPD.^{7,9–11} The risk for PPD has also been found to increase in women with obstetrical and perinatal pregnancy complications or other comorbid conditions, including diabetes, gestational diabetes, hypertension and preeclampsia, cesarean delivery, excessive intrapartum bleeding, and obesity.^{12–15} Other clinical factors that predispose to PPD include poor prenatal care, tobacco use, preterm delivery, and a lack of breastfeeding.^{6,12,16–18} Lastly, according to a large prospective cohort study of more than 700,000 mothers, women with a history of depression are at 20 times higher risk for PPD than those without, after controlling for several PPD risk factors.¹⁴ The cohort included adolescent mothers with an age between 15 and 19, which accounted for only 2 percent (17,823 persons) of the entire study population. Thus, the finding that the history of depression as a significant risk factor for PPD was largely driven by women age 20 or greater.

These same maternal factors, however, may manifest differently among women of younger age at delivery in terms of their prevalence and influence on PPD. This is reflected in a higher PPD rate among adolescents as compared to adult mothers.^{19,20} To the best of our knowledge, studies assessing the relationship between maternal factors and the development of PPD have largely focused on adults or have not been stratified into adult and adolescent groups, thus potentially limiting

our understanding of the role of maternal factors in adolescent PPD.

The objective of this cohort study was to estimate the prevalence of diagnosed PPD and several maternal factors, some of which are well-known risk factors for PPD, among women having their first experience of giving birth and to evaluate whether there are differences in the relationships among various maternal factors and PPD in adolescents (aged 12–19 years) and in adult mothers (aged 20 years or greater). Maternal factors in three broad categories were used in this study, including demographic or socioeconomic factors, psychological factors, and clinical factors.

2 | METHODS

2.1 | Data source and study population

This retrospective cohort study is part of a large project that linked three data sources at the patient level.^{21,22} The three data sources were (1) electronic health records (EHR) from Intermountain Healthcare, (2) EHR from the University of Utah Health, and (3) the Utah Population Database (UPDB). To illustrate, women who had their first live birth between 2008 and 2015 and at least one non-emergency visit within 10 years before their first live birth at either Intermountain Healthcare or the University of Utah Health were linked to Utah birth records available in the UPDB by probabilistic data linkage.²¹ This study was a secondary analysis of the established data and was designed to assess the relationship between LARC and pregnancy outcomes.

The UPDB contains basic demographic information, birth records, death records, and US Census data on Utah residents.²³ Using the linked data, we were able to use in- and outpatient EHR records as well as data available in UPDB for the same woman to measure study variables as described below.

2.2 | Measurement of study variables

2.2.1 | Postpartum depression (PPD)

Postpartum depression was defined as having a primary diagnosis code of major depression in at least one in- or outpatient encounter occurring within 1 year after the

delivery date. The selection of depression-related diagnosis codes was based on Clinical Classification Software (CCS) provided by the Agency for Healthcare Research and Quality (AHRQ). Furthermore, we adapted the CCS groupings by adding additional diagnosis codes indicating depression from the “bipolar disorder” and “other miscellaneous mental conditions” categories to the “depressive disorder” category.²⁴ The final list of diagnosis codes that we used to define postpartum depression is available in Table S1.

2.2.2 | Maternal factors potentially associated with PPD

We referred to a landmark systematic review to look for known PPD risk factors,⁸ and included additional factors that were determined a priori by the authors, who are clinical and/or content experts in maternal child health. Among the known and potential risk factors, we selected those that could be defined operationally in our data set using available data elements or relevant proxy measures. *Demographic and socioeconomic factors* included maternal age, race, ethnicity, educational attainment at delivery, residence at delivery (rural vs. urban), source of payment at delivery (measured by health-insurance type), and marital status at delivery.^{6–12} *Psychological factors* included pre-existing depression and anxiety.^{7,8,14} Finally, *clinical factors* included usage of prenatal care during the first trimester of pregnancy, tobacco use during pregnancy, long-acting reversible contraceptive (LARC) use 1 year before pregnancy, diabetes, hypertension, infections during pregnancy, cesarean delivery, excessive intrapartum bleeding, breastfeeding, preterm birth, and pre-pregnancy body mass index (BMI).^{6,12–18,25–27}

The data sources and definitions used for the maternal factors are shown in Table S1.

With respect to missing data, we replaced the missing value by an observed value using non-stochastic imputation techniques if missing data constituted less than 5% of the adolescent population.^{28,29} In brief, we used the value that accounted for the majority of the study population with the available data as a means of imputing the missing data value. If the missing data were 5% or more, we used the indicator method, which involved creating an additional “missing” category for the variable. Subsequently, the correlations between study variables were examined by correlation analysis.³⁰ If any significant correlations were found, we considered combining all the correlated factors together to make a composite variable, or we retained only a variable with a large enough sample size and a better univariate association with PPD, excluding the others.

2.3 | Statistical analysis

Because this study aimed to explore differences in the association (direction and magnitude) of maternal factors for PPD between adolescent and adult mothers, the cohort was stratified and analyzed in two separate models depending on maternal age at first childbirth. Adolescents were defined as persons aged 12–19 years, and adults were defined as persons aged 20 years or older, according to the Centers for Disease Control and Prevention (CDC) definitions.³¹ Multivariable logistic regression was used to evaluate the associations between maternal factors and PPD, and estimates were reported as odds ratios (OR) with 95% confidence intervals (CIs).

Analyses were run using SAS software 9.4 (Copyright, SAS Institute Inc. SAS and all other SAS Institute Inc. product or service names are registered trademarks or trademarks of SAS Institute Inc.). The study was reviewed and approved by the University of Utah Institutional Review Board (IRB_00103461).

3 | RESULTS

Among the 61,226 women included in the study cohort of first-time mothers, 10.5% ($N=6435$) were adolescents, and 89.5% ($N=54,791$) were adults. The average maternal ages were 25.1 years (standard deviation, $SD=4.76$) for the pooled cohort and 18.0 years ($SD=1.12$) and 25.9 years ($SD=4.31$) for the adolescent and adult cohorts, respectively.

The patient characteristics for the age-stratified cohorts are in Table 1. Overall, 4.04% ($N=2475$) of the mothers in the study population had PPD, defined as having a depression-related diagnostic code on one or more healthcare encounters within 365 days of the date of delivery. The PPD rates were significantly different between adolescents and adults. Among adolescents, 6.1% ($N=391$) had a PPD-related diagnostic code, compared with only 3.8% ($N=2084$) of the adult mothers. Similarly, the adolescent cohort had a greater proportion of mothers with a putative risk factor for PPD than the adult cohort (Non-White race, Hispanic ethnicity, rurality, social insurance or no insurance, unmarried, absence of prenatal visit in the first trimester, recent tobacco use, infections such as gonorrhea, syphilis, chlamydia, or hepatitis B or C during pregnancy, no documented records of breastfeeding, preterm birth, underweight defined by a BMI of 18.5 or less, had depression and anxiety before delivery). The proportion of mothers was higher among adults than adolescents for the remaining PPD (putative) risk factors, including history of LARC use, diabetes, hypertension, cesarean delivery, intrapartum bleeding, or a BMI of 25 or greater.

TABLE 1 Characteristics of the study population and age-stratified cohorts.

Variables	All (N = 61,226)	Age-stratified	
		Adult (N = 54,791)	Adolescent (N = 6435)
Maternal age, mean (SD)	25.11 (4.76)	25.94 (4.31)	18.01 (1.12)
Maternal race, N(%)			
White	57,982 (94.70%)	52,024 (94.95%)	5958 (92.59%)
Non-white	3244 (5.30%)	2767 (5.05%)	477 (7.41%)
Maternal ethnicity, N(%)			
Hispanic	10,504 (17.16%)	8015 (14.63%)	2489 (38.68%)
Non-Hispanic	50,722 (82.84%)	46,776 (85.37%)	3946 (61.32%)
Maternal education attainment at delivery, N(%)			
No high school degree	6771 (11.06%)	3135 (5.72%)	3636 (56.50%)
High school graduate	12,216 (19.95%)	10,031 (18.31%)	2185 (33.95%)
Some college	21,665 (35.39%)	21,051 (38.42%)	614 (9.54%)
College graduate	20,574 (33.60%)	20,574 (37.55%)	0 (0.0%)
Residence at delivery, N(%)			
Rural	5860 (9.57%)	4975 (9.08%)	885 (13.75%)
Urban	55,366 (90.43%)	49,816 (90.92%)	5550 (86.25%)
Source of payment for delivery, N(%)			
Social insurance	15,097 (24.66%)	11,676 (21.31%)	3421 (53.16%)
Self-pay, uninsured, or unknown	12,435 (20.31%)	10,609 (19.36%)	1826 (28.38%)
Private insurance	33,694 (55.03%)	32,506 (59.33%)	1188 (18.46%)
Marital status at delivery, N(%)			
Unmarried	16,256 (26.55%)	10,944 (19.97%)	5312 (82.55%)
Married	44,970 (73.45%)	43,847 (80.03%)	1123 (17.45%)
First trimester prenatal visit, N(%)			
Yes	50,080 (81.80%)	45,797 (83.58%)	4283 (66.56%)
No	11,146 (18.20%)	8994 (16.42%)	2152 (33.44%)
Recent tobacco use, N(%)			
Yes	3773 (6.16%)	3068 (5.6%)	705 (10.96%)
No	57,453 (93.84%)	51,723 (94.40%)	5730 (89.04%)
Long-acting reversible contraceptive use, N(%)			
Yes	1044 (1.71%)	1006 (1.84%)	38 (0.59%)
No	60,182 (98.29%)	53,785 (98.16%)	6397 (99.41%)
Pre-existing or gestational diabetes, N(%)			
Yes	2055 (3.36%)	1965 (3.59%)	90 (1.45%)
No	59,171 (96.64%)	52,826 (96.41%)	6345 (98.60%)
Hypertension before or during pregnancy, N(%)			
Yes	5402 (8.82%)	4991 (9.11%)	411 (6.39%)
No	55,824 (91.18%)	49,800 (90.89%)	6024 (93.61%)
Infections during pregnancy, N(%)			
Yes	966 (1.58%)	49 (1.18%)	317 (4.93%)
No	60,260 (98.42%)	54,742 (99.91%)	6118 (95.07%)
Cesarean delivery, N(%)			
Yes	13,070 (21.35%)	12,113 (22.11%)	957 (14.87%)
No	48,156 (78.65%)	42,678 (77.89%)	5478 (85.13%)

(Continues)

TABLE 1 (Continued)

Variables	All (N = 61,226)	Age-stratified	
		Adult (N = 54,791)	Adolescent (N = 6435)
Excessive intrapartum bleeding, N(%)			
Yes	2105 (3.44%)	1941 (3.54%)	164 (2.55%)
No	59,121 (96.56%)	52,850 (96.46%)	6271 (97.45%)
Breastfeeding, N(%)			
Yes	44,576 (72.81%)	40,719 (74.32%)	3857 (59.94%)
No	16,650 (27.19%)	14,072 (25.68%)	2578 (40.06%)
Preterm birth, N(%)			
Yes	4572 (7.47%)	3972 (7.25%)	600 (9.32%)
No	56,654 (92.53%)	50,819 (92.75%)	5835 (90.68%)
Pre-pregnancy body mass index, N(%)			
BMI ≤ 18.5	3510 (5.73%)	2836 (5.18%)	674 (10.47%)
25.0 ≤ BMI < 30	12,464 (20.36%)	11,361 (20.74%)	1103 (17.14%)
18.5 ≤ BMI < 25	36,318 (59.32%)	32,240 (58.84%)	4078 (63.37%)
BMI ≥ 30	8934 (14.59%)	8354 (15.25%)	580 (9.01%)
Depression before delivery, N(%)			
No	57,033 (93.15%)	51,239 (93.52%)	5794 (90.04%)
Yes	4193 (6.85%)	3552 (6.48%)	641 (9.96%)
Before and during pregnancy	755 (1.23%)	641 (1.17%)	114 (1.77%)
Before pregnancy only	1256 (2.05%)	1006 (1.84%)	250 (3.89%)
During pregnancy only	2182 (3.56%)	1905 (3.48%)	277 (4.30%)
Anxiety before delivery, N(%)			
Yes	471 (0.77%)	402 (0.74%)	69 (1.07%)
No	60,755 (99.23%)	54,389 (99.27%)	6366 (98.93%)
Postpartum depression, N(%)			
Yes	2475 (4.04%)	2084 (3.80%)	391 (6.08%)
No	58,751 (95.96%)	52,707 (96.20%)	6044 (93.92%)

Abbreviations: BMI, body mass index; SD, standard deviation.

The multivariable logistic regression analyses in the two stratified cohorts identified some notable similarities and differences in terms of the direction and magnitude of associations between maternal factors and PPD. These are presented as ORs with 95% CIs in Table 2.

3.1 | Demographic or socioeconomic factors

Maternal age at delivery was associated with PPD in both adolescent and adult mothers, but in a different direction of association (OR: 1.03 [95% CI: 1.02–1.04] in adults; OR: 0.88 [95% CI: 0.79–0.99] in adolescents). Specifically, younger age among adolescents and older age among adults were significantly associated with PPD. Rural residence was significantly associated with lower odds of PPD diagnosis compared with living in urban areas among adolescent

mothers (OR: 0.59 [95% CI: 0.40–0.86]) but not among adult mothers (OR: 0.94 [95% CI: 0.79–1.12]). Both adolescent and adult women who were self-paying, uninsured, or who had missing insurance data were less likely to have a PPD diagnosis than women who were privately insured (OR: 0.73 [95% CI: 0.63–0.86] in adults; OR: 0.58 [95% CI: 0.42–0.82] in adolescents). More than 20% of adult mothers had no education beyond high school, and these women had a higher risk of PPD than women who attended college (no high school degree OR: 1.31 [95% CI: 1.08–1.59]; high school graduate OR: 1.17 [95% CI: 1.02–1.33]).

3.2 | Clinical factors

Both tobacco use during pregnancy (OR: 1.28 [95% CI: 1.02–1.31] in adults; OR: 1.86 [95% CI: 1.40–2.47] in adolescents) and, while rare, pre-pregnancy LARC use

TABLE 2 Associations between candidate risk factors and postpartum depression in age-stratified cohorts.

Variables	Odds Ratio (95% Confidence Interval)	
	Adults	Adolescents
Demographic/socioeconomic factors		
Maternal age	1.03 (1.02–1.04)	0.88 (0.79–0.99)
Race, White (ref=Non-white)	0.97 (0.79–1.21)	1.16 (0.78–1.72)
Ethnicity, Hispanic (ref=Non-Hispanic)	0.73 (0.63–0.85)	0.79 (0.62–1.01)
Education attainment at delivery (ref=Some College)		
No High School Degree	1.31 (1.08–1.59)	1.00 (0.78–1.25)
High School Graduate	1.17 (1.02–1.33)	0.96 (0.64–1.46)
College Graduate	0.91 (0.80–1.03)	Not applicable
Rural (ref=Urban)	0.94 (0.79–1.12)	0.59 (0.40–0.86)
Source of payment for delivery (ref=Private Insurance)		
Social Insurance	1.18 (1.04–1.34)	0.81 (0.62–1.06)
Others (Self-pay, uninsured, or missing)	0.73 (0.63–0.86)	0.58 (0.42–0.82)
Unmarried (ref=Married)	1.29 (1.13–1.46)	1.04 (0.76–1.43)
Clinical factors		
No prenatal visit during first trimester	1.15 (1.02–1.31)	0.99 (0.78–1.25)
Recent tobacco use	1.28 (1.12–1.41)	1.86 (1.40–2.47)
Pre-pregnancy LARC use	1.50 (1.16–1.94)	4.37 (1.94–9.86)
Pre-existing or gestational diabetes	1.07 (0.85–1.33)	0.83 (0.35–1.96)
Pre-existing or pregnancy hypertension	1.21 (1.04–1.40)	1.26 (0.85–1.87)
Infections during pregnancy	1.28 (0.92–1.79)	0.94 (0.57–1.53)
Cesarean delivery	1.17 (1.05–1.31)	1.20 (0.90–1.60)
Excessive intrapartum bleeding	1.11 (0.86–1.44)	0.98 (0.48–1.98)
No breastfeeding	1.16 (1.03–1.31)	1.22 (0.96–1.55)
Preterm birth	1.19 (1.12–1.40)	1.23 (0.88–1.73)
Pre-pregnancy body mass index (ref=Normal weight, 18.5≤BMI<25)		
Underweight, BMI≤18.5	1.10 (0.89–1.36)	0.81 (0.55–1.20)
Overweight, 25.0≤BMI<30	1.20 (1.06–1.35)	1.08 (0.81–1.45)
Obese, BMI≥30	1.36 (1.20–1.54)	1.27 (0.89–1.82)
Psychological factors		
Depression before delivery (ref=no depression before delivery)		
Both pre- and during pregnancy	20.81 (17.42–24.86)	9.20 (5.93–14.27)
Pre-pregnancy only	8.82 (7.43–10.47)	3.49 (2.38–5.11)
During pregnancy only	8.60 (7.53–9.81)	5.70 (4.13–7.85)
Anxiety before delivery	6.49 (5.12–8.24)	4.19 (2.37–7.14)

Abbreviations: BMI, body mass index; LARC, long-acting reversible contraceptives; ref, reference.

(OR: 1.50 [95% CI: 1.16–1.94] in adults; OR: 4.37 [95% CI: 1.94–9.86] in adolescents) were significantly associated with PPD in both adolescent and adult mothers. Delayed prenatal care initiation was a significant risk factor for PPD among adult mothers (OR: 1.15 [95% CI: 1.02–1.31]), but the association was not significant in adolescent mothers (OR: 0.99 [95% CI: 0.78–1.25]). None of the remaining clinical factors were found to be

significantly associated with adolescent PPD, although some of them were significant in adult mothers, such as pre-pregnancy BMI of 25 or greater (OR of BMI between 25 and 30: 1.20 [95% CI: 1.06–1.35]; OR of BMI 30 or greater: 1.36 [95% CI: 1.20–1.54]), hypertension (OR: 1.21 [95% CI: 1.04–1.40]), cesarean delivery (OR: 1.17 [95% CI: 1.05–1.31]), and no breastfeeding at discharge (OR: 1.16 [95% CI: 1.03–1.31]).

3.3 | Psychological factors

In general, the history of a diagnosis of depression was significantly associated with PPD, with ORs among adults of 8.60 and 20.81 and among adolescents of 3.49 and 9.20 for pre-pregnancy and intrapartum depression, respectively. Similarly, a diagnosis of anxiety before or during pregnancy was significantly associated with PPD (OR: 6.49 [95% CI: 5.12–8.24] in adults, OR: 4.19 [95% CI: 2.37–7.14] in adolescents; See Table 2).

4 | DISCUSSION

We found that the PPD rate was 1.5 times higher among adolescent mothers (6.1%) than adult mothers (3.8%). Notably, the proportion of birthing people with clinically identified PPD in our study (4.0%) was lower than the self-reported rate of postpartum depressive symptoms in state data (12.5%), as estimated by the 2017 Pregnancy Risk Assessment Monitoring System (PRAMS) depression screening questions.³² This study defined PPD using diagnostic codes, while the PRAMS survey uses a PHQ-2 depression screen to identify the prevalence of depressive symptoms.³³ Our study findings reflect a clinically meaningful difference in diagnosed PPD prevalence between adult and adolescent moms, supported by existing literature, which has consistently found a higher PPD rate in women aged under 20 than in adults.^{34,35}

Our analyses identified common maternal factors potentially associated with PPD between adults and adolescents, as well as unique associations in adults and adolescents. This study adds to the literature by providing data on maternal factors associated with PPD that are more prevalent among adolescents than adult mothers, highlighting the need to address and understand maternal factors that may contribute to a PPD vulnerability among adolescents. Clinicians should consider these factors in adolescent-focused PPD education, screening, and referral.

4.1 | Clinical factors and PPD

Adolescent mothers have a higher prevalence of PPD compared with adult mothers. It is notable that adolescent mothers were twice as likely as adult mothers to delay prenatal care initiation (33.4% vs. 16.42%, p -value < 0.001) in our study.³⁶ Early prenatal care provides an opportunity to expand patient education throughout the pregnancy period about counseling for smoking cessation, perinatal depression, and screening for infections and rubella immunity.³⁶ This was reflected in our finding with the

higher prevalence of lack of prenatal care in adolescents as well as the higher prevalence of tobacco use (11.0% vs. 5.6%, p -value < 0.001) and infections (4.9% vs. 1.2%, p -value < 0.001) during pregnancy compared with adults. Lack of early prenatal care did not significantly increase adolescent PPD risk (OR: 0.99 [0.78–1.25]). However, the lack of prenatal care during the first trimester was associated with higher rates of adult PPD. This may indicate a disparity between adult and adolescent mothers in terms of the effectiveness of early prenatal care on PPD development. Implementing effective prenatal care in the first trimester should be emphasized as an opportunity for practitioners to intervene on risk factors that have significant associations with PPD, such as smoking, pre-existing mental health conditions, and other perinatal complications, and to engage patients in mitigating these risk factors. A recent randomized clinical trial also supports the idea in that it found prenatal care can reduce the risk of PPD in primiparous adolescent mothers.³⁷

4.2 | Demographic or socioeconomic factors and PPD

In line with previous study results, increasing maternal age was associated with higher odds of PPD among adult mothers with an OR of 1.03 (95% CI: 1.02–1.04), but, among adolescents, younger adolescents had a higher risk for PPD with an OR of 0.88 (95% CI: 0.79–0.99).³⁸ This finding differs from CDC reports of depression prevalence by age group in the US, for which the odds of depression continue to increase with age from age 12 until the 60s in females, suggesting that reproductive events at a younger age play some role in the etiology of PPD, differentiated from other possible factors that have long been associated with depression in general.^{39,40}

Our finding that adolescents living in rural areas had lower odds of experiencing PPD warrants furthermore study. While this finding aligns with previous research demonstrating lower rates of PPD in rural women compared with their urban counterparts, it raises important questions with respect to the influence of limited access to maternity care services in rural communities.^{41,42} Studies have shown that rural postpartum women face barriers in seeking healthcare services, including depression screening and access to mental health care.⁴³ With the increasing challenges posed by hospital and obstetric department closures as well as workforce shortages in rural areas, the availability and quality of maternal health services may be compromised, leading to a potential underestimation of PPD prevalence in these communities.⁴⁴ Consequently, addressing the specific needs and challenges of rural areas becomes essential

to improving maternal outcomes and enhancing mental health support for rural women. Furthermore, future research should be designed to better understand the complexities and age-related disparities (i.e., adolescent vs. adult) in identifying and managing PPD within the context of limited access to maternity care in rural communities.

4.3 | Psychological factors and PPD

Notably, a history of depression or anxiety before delivery was strongly associated with PPD. These were the factors with the strongest associations with PPD for both groups of mothers, but the magnitude was larger among adolescent mothers. This suggests depression was likely to continue, recur, or worsen during the postpartum period. The strong association between pre-existing depression or anxiety and PPD is consistent with previous studies. Pre-pregnancy anxiety and depression have been associated with poor contraceptive behavior, earlier age at first intercourse, more sexual partners, ambivalent pregnancy desire, lack of contraceptive knowledge, and low self-efficacy, which may also be related to PPD.^{45,46} This furthermore highlights the importance of universal screening for depression and anxiety and treatment initiation among those with positive screening results,⁴⁷ in alignment with recommendations from the American College of Obstetricians and Gynecologists and the Women's Preventive Services Initiative.^{48,49}

4.4 | Strengths and limitations

A major strength of our study is the use of population-based data to understand established maternal factors associated with PPD between adults and adolescents separately. Also, our study included more than 6000 adolescent mothers, making it one of the largest adolescent-mother cohorts in the literature. Last, using linked data from clinical encounters and public health or administrative data surrounding pregnancy allowed us to study both clinical and non-clinical factors associated with PPD in the large cohort. Of note, the two healthcare systems represented in the study account for approximately 85% of healthcare in the state of Utah; thus, the cohort represented the vast majority of births in the state.

Our study also has some limitations. One limitation involves the misclassification of PPD and maternal factors potentially associated with PPD. Defining PPD by way of diagnostic codes only identifies clinical encounters where a depression-related ICD-9 or 10 code was recorded; this excludes women who were not screened or whose visit

included PPD-related care that was not documented with a diagnostic code. Women without insurance or residing in rural areas may have had less access to healthcare and therefore been less likely to receive a PPD diagnosis or PPD-related care, making it appear that lack of insurance itself was associated with a lower risk of "diagnosed" PPD. Because diagnosis records in the EHR may not be fully representative of PPD prevalence, some mild PPDs may have fallen below the diagnostic certainty level, resulting in low diagnostic sensitivity. However, this approach has high specificity and therefore minimizes the likelihood of including patients without an accurate ICD diagnosis. Similarly, clinical records outside the two health systems were not available; thus, variable definitions that used the mother's EHR were subject to misclassifying patients having certain maternal factors as not having them. The affected variables are the history of depression and anxiety before conception and recent LARC use (Table S1). Under these conditions, the magnitude of associations between the maternal factors and PPD could have been underestimated. Given that we cannot estimate the rate of misclassified patients for the risk factors in this study population, the extent of underestimation remains unknown. Similarly, while the 10 years of historical data before a birth represent a strength of the cohort, a related limitation is that younger adolescents have fewer years of life during which to assess the existence of maternal factors before their first birth. However, improved identification of depression has occurred as practitioners have begun to screen more frequently during postpartum and pediatric appointments; the American Academy of Pediatrics recommended routine screening for postpartum depression in 2010,⁵⁰ and screening in pediatric settings has been shown to increase the identification of women suffering from PPD.^{51,52}

In addition, while we were able to measure several important factors associated with PPD in our analysis, unintended pregnancy, which is known to be one of the strongest risk factors for PPD, was not considered in the analysis.⁵³ It is possible that the observed associations between measured maternal factors and PPD could have been an artifact association in the absence of controlling for unintended pregnancy. However, the candidate maternal factors included in our analysis are well-established risk factors for PPD; thus, the possibility should generally be considered ruled out.

5 | CONCLUSIONS

Despite study limitations, we identified a higher PPD rate in adolescent mothers compared with adults as well as several maternal factors associated with PPD that are

more prevalent in nulliparous adolescents compared with adults in a large population-based study. Identifying, and where possible, addressing these risk factors in new mothers may affect PPD outcomes. Routine prenatal depression and anxiety screening, as well as PPD screening, is critical to identifying and referring women who need care. Depression and anxiety surveillance during preconception and prenatal appointments should be used to identify worsening pregnancy-related symptoms that could ultimately manifest as PPD. Women of all ages (including adolescents) should be routinely screened for depression and anxiety before, during, and after pregnancy—a suggestion that is in alignment with national guidelines.^{48,49} In states that have extended Medicaid coverage during the 1-year postpartum period, women avail themselves of more Medicaid-financed outpatient visits during the postpartum year.⁵⁴ These visits provide opportunities for PPD screening and treatment. In states without expanded coverage, women become uninsured after 60 days, which restricts their ability to seek care for conditions such as postpartum depression, which often emerges after 60 days postpartum. Thus, Medicaid coverage expansion throughout the United States will likely increase the number of women who are diagnosed and treated for PPD.

In addition, adults and adolescents who enter pregnancy with anxiety or depression or who develop anxiety or depression during pregnancy should be counseled about treatment options, weighing the risks and benefits.⁵⁵ Adolescents may be at increased risk of PPD compared with adult women and may benefit from enhanced support and care in the postpartum period. Finally, additional research on risk or protective factors unique to adolescents may provide insights into the mechanisms behind these associations and information useful for developing and tailoring PPD interventions for this vulnerable group.

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DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the University of Utah. Restrictions apply to the availability of these data, which were used under license for this study. Data are available from the author(s) with the permission of the University of Utah.

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